Adaptive Experimental Design Using Shrinkage Estimators

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Outline

1 Background and Motivation

2 Estimators and Risk

3 Oracle Designs

4 Adaptation & Inference

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  - assign website visitors to a different UI as they arrive
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- **Goal:** Obtain best estimates of *all* treatment effects (vs. the control) at the end of the trial

Building Blocks

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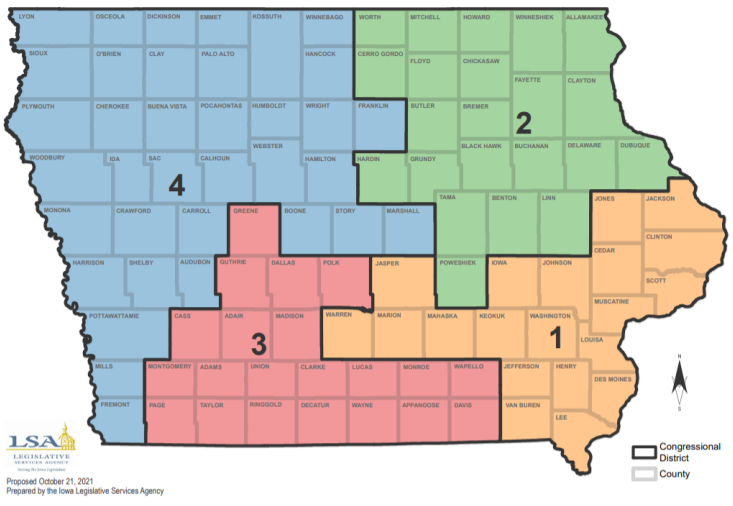
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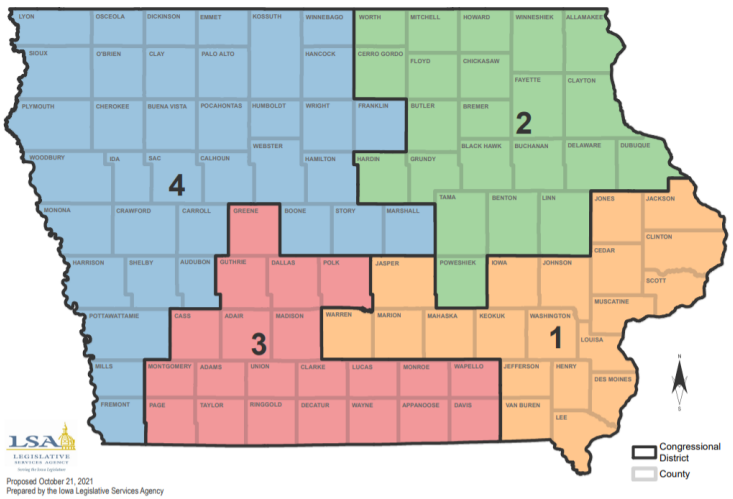
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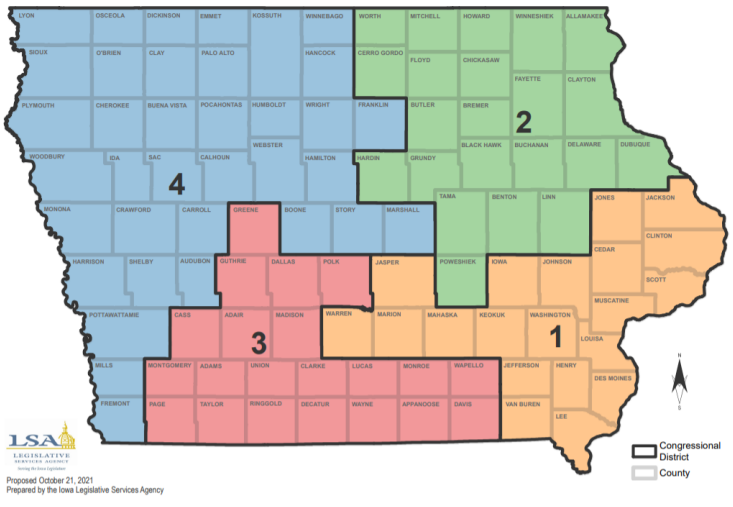


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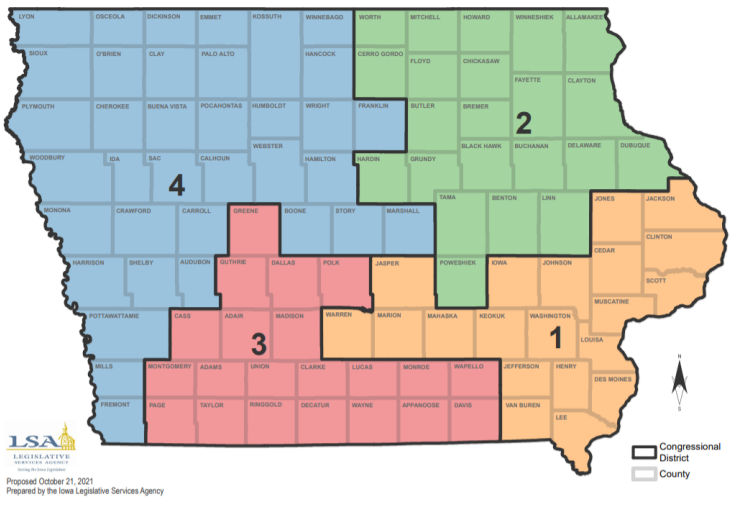
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  - Variants of this approach appear in Blackwell et al. (2023), Dai et al. (2023), Zhao (2023), Li and Owen (2024) and Cai and Rafi (2024)

Idea #2: Shrinkage Estimation

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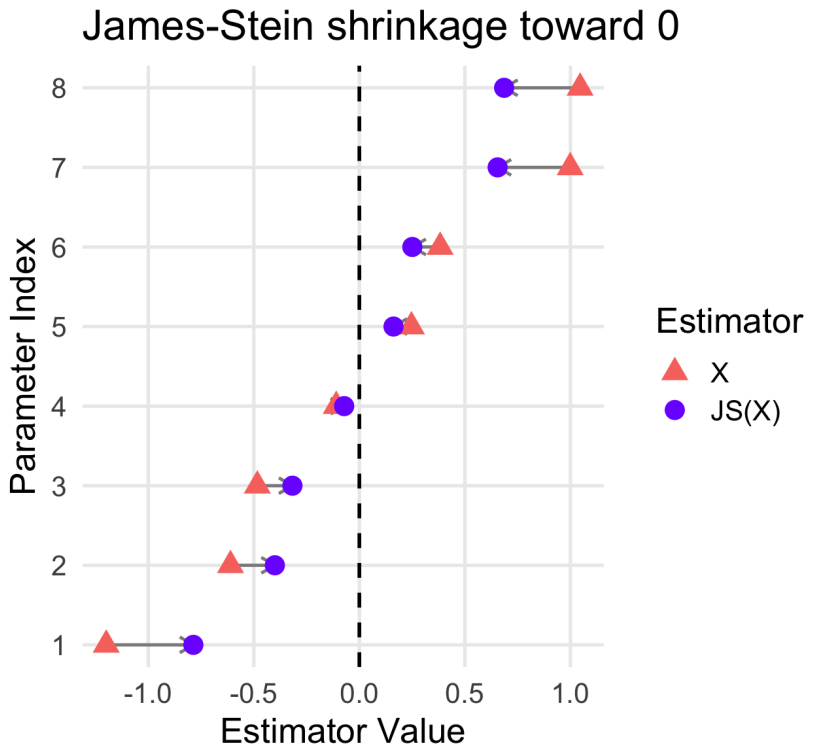
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Idea #2: Shrinkage Estimation

Recent work uses shrinkage estimation in **causal inference**:

- Synthesizing observational and experimental causal estimates (Chen et al., 2015; Rosenman et al., 2023; Lin et al., 2025)
- Improved precision in online experiments (Dimmery et al., 2019; Deng et al., 2021)

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Simulations demonstrate significant performance improvements, especially in low signal-to-noise ratio regime!

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- Seek to estimate vector of causal effects $\boldsymbol{\tau}$, where

$$\begin{aligned} \boldsymbol{\tau} &= (\tau_1, \ldots, \tau_K) \\ &= (\mu_1 - \mu_0, \ldots, \mu_K - \mu_0) \in \mathbb{R}^K \end{aligned}$$

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- In a standard trial, with large n_k, approx. $\boxed{\hat{\boldsymbol{\tau}} \sim \mathcal{N}(\boldsymbol{\tau}, \boldsymbol{\Sigma})}$,

$$\boldsymbol{\Sigma} = \begin{pmatrix} \frac{V_1}{n_1} + \frac{V_0}{n_0} & \frac{V_0}{n_0} & \cdots & \frac{V_0}{n_0} \\ \frac{V_0}{n_0} & \frac{V_2}{n_2} + \frac{V_0}{n_0} & \cdots & \frac{V_0}{n_0} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{V_0}{n_0} & \frac{V_0}{n_0} & \cdots & \frac{V_K}{n_K} + \frac{V_0}{n_0} \end{pmatrix}.$$

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- With diff-in-means estimator $\hat{\boldsymbol{\tau}}$,

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$$= \underbrace{\frac{KV_0}{n_0}}_{\substack{\text{"excess" contribution} \\ \text{of control}}} + \sum_{k=1}^{K} \frac{V_k}{n_k}.$$

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  - Risk $\mathcal{R}_{\boldsymbol{\delta}}(\boldsymbol{\tau}, \boldsymbol{\Sigma})$ will depend on both $\boldsymbol{\tau}$ and $\boldsymbol{\Sigma}$.

Candidate Estimators

Consider shrinkage estimators that work with *dependent* and *heteroscedastic* data.

- Bock's estimator (Bock, 1975)

$$\boldsymbol{\delta_B} = \left(1 - \frac{\tilde{p} - 2}{\hat{\boldsymbol{\tau}}^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \hat{\boldsymbol{\tau}}}\right) \hat{\boldsymbol{\tau}} \qquad \text{where} \qquad \tilde{p} = \frac{\mathrm{Tr}(\boldsymbol{\Sigma})}{\lambda_{\max}(\boldsymbol{\Sigma})}$$

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1. Compute an unbiased risk estimate (modified version of SURE; Stein, 1981)

Step 1: Unbiased Risk Estimate

Corollary (Estimator Risk)

Suppose we have $\hat{\boldsymbol{\tau}} \sim \mathcal{N}(\boldsymbol{\tau}, \boldsymbol{\Sigma}) \in \mathbb{R}^K$ *for general* $\boldsymbol{\Sigma}$. *Consider*

$$\boldsymbol{\kappa}(\hat{\boldsymbol{\tau}}) = \hat{\boldsymbol{\tau}} - \boldsymbol{g}(\hat{\boldsymbol{\tau}})$$

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Upshot: slight modification of classical SURE to account for heteroscedasticity and dependence across causal estimates.

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$$\mathcal{R}_{\boldsymbol{\delta_S}}(\boldsymbol{\tau}, \boldsymbol{\Sigma}) = \mathbb{E}_{\mathcal{F}}\left(||\boldsymbol{\delta_S} - \boldsymbol{\tau}||_2^2\right) = \mathbb{E}_{\mathcal{F}}\left(\sum_{k=1}^{K}(\delta_{sk} - \tau_k)^2\right).$$

1. Compute an unbiased risk estimate (modified version of SURE; Stein, 1981)

$$\mathcal{R}_{\boldsymbol{\delta_S}}(\boldsymbol{\tau}, \boldsymbol{\Sigma}) = \mathrm{Tr}(\boldsymbol{\Sigma}) + \mathrm{Tr}(\boldsymbol{\Sigma})\underbrace{\mathbb{E}\left(\frac{\hat{\boldsymbol{\tau}}^{\mathsf{T}}\left(4\boldsymbol{\Sigma} - \mathrm{Tr}(\boldsymbol{\Sigma})\boldsymbol{I}_K\right)\hat{\boldsymbol{\tau}}}{||\hat{\boldsymbol{\tau}}||_2^4}\right)}_{\substack{\text{expectation of ratio of} \\ \text{Gaussian quadratic forms}}}$$

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2. Implement expectation as a *univariate* numerical integral via the method of Bao and Kan (2013)

Outline

1 Background and Motivation

2 Estimators and Risk

3 Oracle Designs

4 Adaptation & Inference

Oracle Setting

- With a way to compute risks of shrinkage estimators, we can optimize experimental designs to minimize these quantities!

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  - Values of μ_k, V_k are known for $k = 0, \ldots, K$
  - Allocate fixed sample size n across the $K + 1$ arms
- Let's start by optimizing allocation over risk of $\hat{\boldsymbol{\tau}}$:

$$\mathcal{R}_{\hat{\boldsymbol{\tau}}}(\boldsymbol{\tau}, \boldsymbol{\Sigma}) = \mathrm{Tr}(\boldsymbol{\Sigma}) = \underbrace{\frac{KV_0}{n_0}}_{\substack{\text{“excess” contribution} \\ \text{of control}}} + \sum_{k=1}^{K} \frac{V_k}{n_k}.$$

(Oracle) Neyman Allocation

Risk-minimizing allocations is solution to Problem 1:

$$\begin{aligned} \min_{n_0,\ldots,n_K} \quad & \frac{KV_0}{n_0} + \sum_{k=1}^{K} \frac{V_k}{n_k} \\ \text{subject to} \quad & \sum_{k=0}^{K} n_k = n, \\ & 0 < n_0, \ldots, n_K. \end{aligned} \tag{1}$$

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Solution works out to be

$$n_0^\star = \frac{n\sqrt{KV_0}}{\sqrt{KV_0} + \sum_{\ell=1}^{K} \sqrt{V_\ell}}, \qquad \text{and}$$

$$n_k^\star = \frac{n\sqrt{V_k}}{\sqrt{KV_0} + \sum_{\ell=1}^{K} \sqrt{V_\ell}} \qquad \text{for } k = 1, \ldots, K.$$

What is the risk-minimizing design?

Risk-minimizing allocation for shrinker $\boldsymbol{\delta_S}$ is solution to Problem 2:

$$\begin{aligned} \min_{n_0,\ldots,n_K} \quad & \mathrm{Tr}(\boldsymbol{\Sigma}) + \mathrm{Tr}(\boldsymbol{\Sigma})\mathbb{E}\left(\frac{\hat{\boldsymbol{\tau}}^{\mathsf{T}}\left(4\boldsymbol{\Sigma} - \mathrm{Tr}(\boldsymbol{\Sigma})\boldsymbol{I}_K\right)\hat{\boldsymbol{\tau}}}{||\hat{\boldsymbol{\tau}}||_2^4}\right) \\ \text{subject to} \quad & \sum_{k=0}^{K} n_k = n, \\ & 0 < n_0, \ldots, n_K. \end{aligned} \tag{2}$$

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- Not convex $\Longrightarrow$ hard to solve globally!

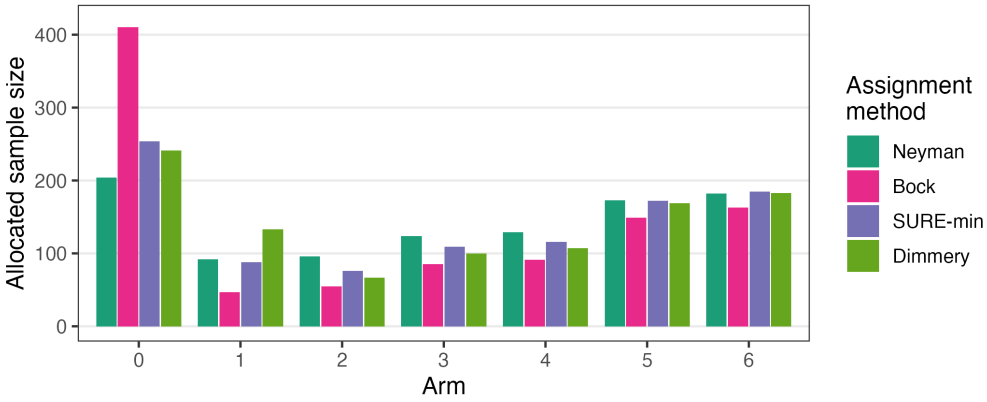
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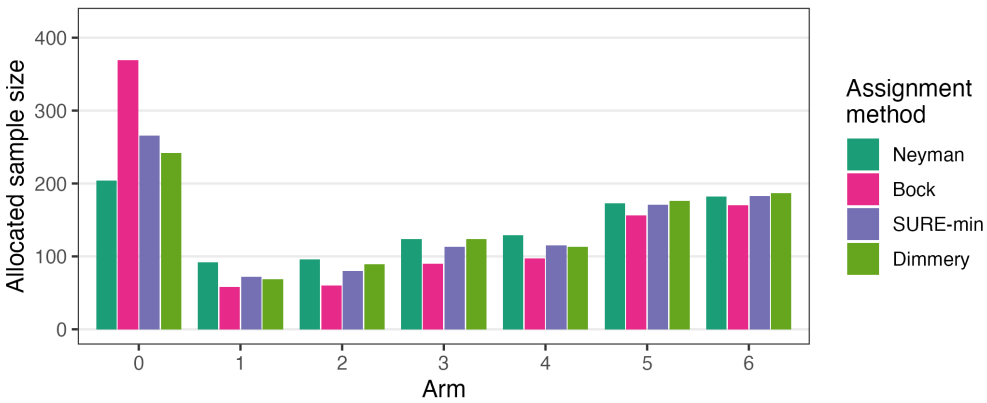
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- Not convex $\Longrightarrow$ hard to solve globally!
- **Greedy minimization**: start at Neyman allocation; swap units across arms until no further improvements possible

Simulated Results: Low Signal-to-Noise Ratio (SNR)



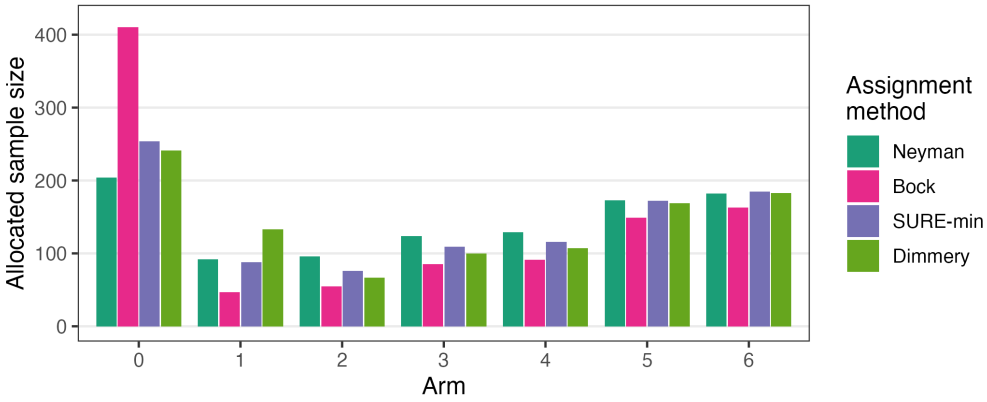
Sparse signal; low SNR



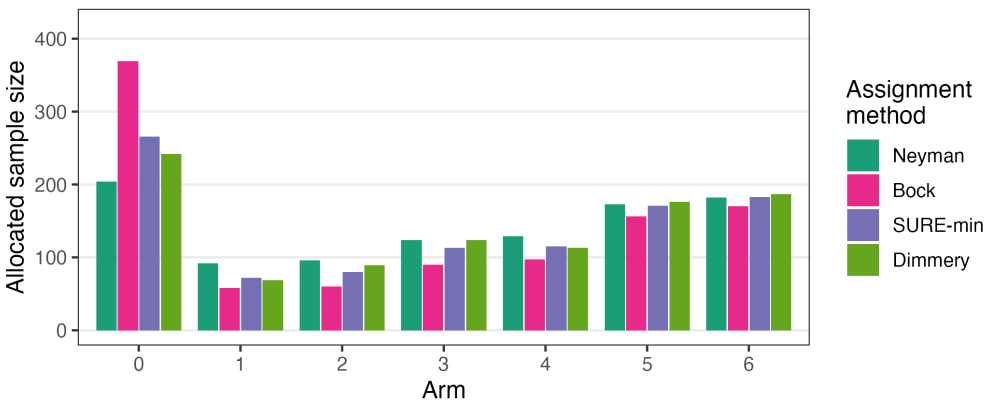
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- Simulated allocations for $K = 6$
- Heteroscedastic, $n = 1,000$

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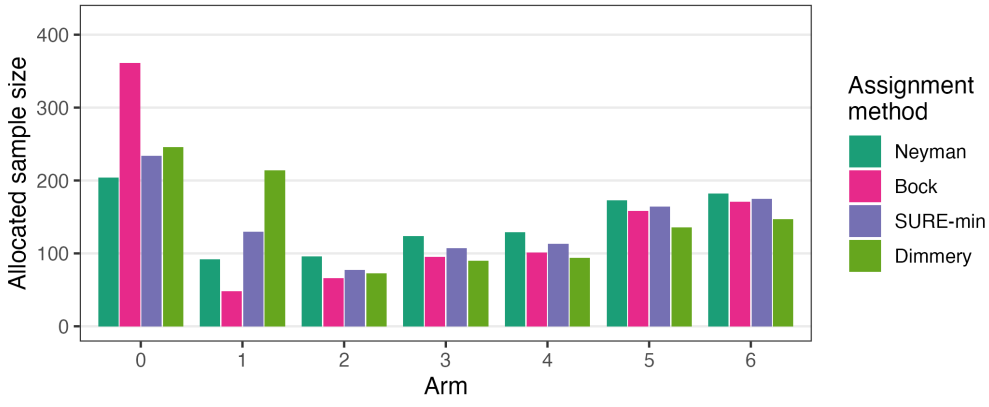
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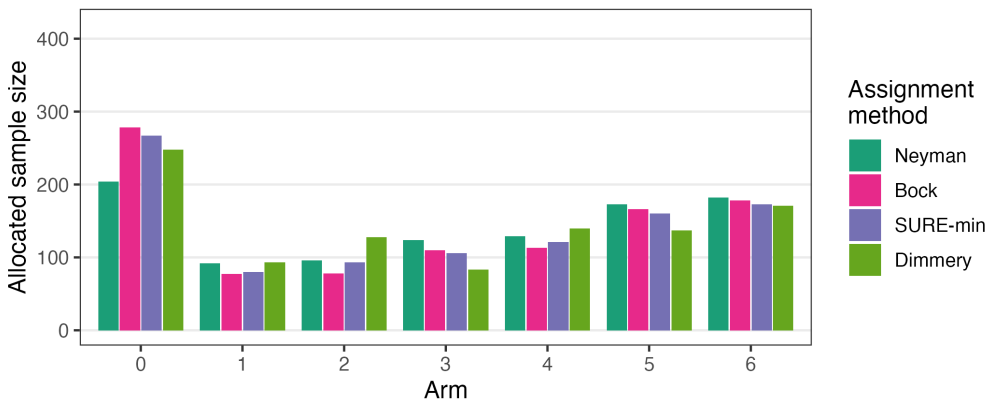
Dense signal; low SNR

- Simulated allocations for $K = 6$
- Heteroscedastic, $n = 1,000$
- Shrinkers allocate more units to control than Neyman
- Sparse signal: $\boldsymbol{\delta_D}, \boldsymbol{\delta_S}$ reallocate more units to strong-effect arm

Simulated Results: High SNR



Sparse signal; high SNR



Dense signal; high SNR

- Stronger signal $\rightarrow$ less reallocation to control
- More oversampling of high-signal arm by δ_D, δ_S

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Adaptivity: Key Ideas

Upshot: With online estimates of $\boldsymbol{\Sigma}$ and $\boldsymbol{\tau}$, can estimate $\mathcal{R}_{\boldsymbol{\delta}}(\boldsymbol{\tau}, \boldsymbol{\Sigma})$ for any desired estimator $\boldsymbol{\delta}$

$\longrightarrow$ adapt future assignments to minimize estimated risk

Adaptivity: Key Ideas

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$\longrightarrow$ adapt future assignments to minimize estimated risk

1. **Greedy algorithm:** for each new arrival i, choose whichever arm $k = 0, 1, \ldots, K$ minimizes current risk estimate.
2. **Batch update:** minimize treatment probabilities for each batch over current risk estimate. Apply IPW weights.
3. **Gradient update:** iterative update of treatment probabilities via risk gradient step, a la Dai et al. (2023). IPW weights.
4. ...

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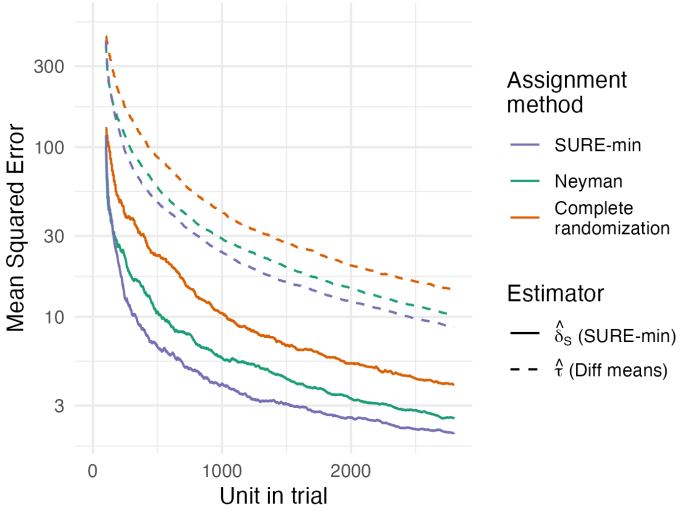
Rcpp Implementation

To achieve required speeds, need extremely fast risk computation

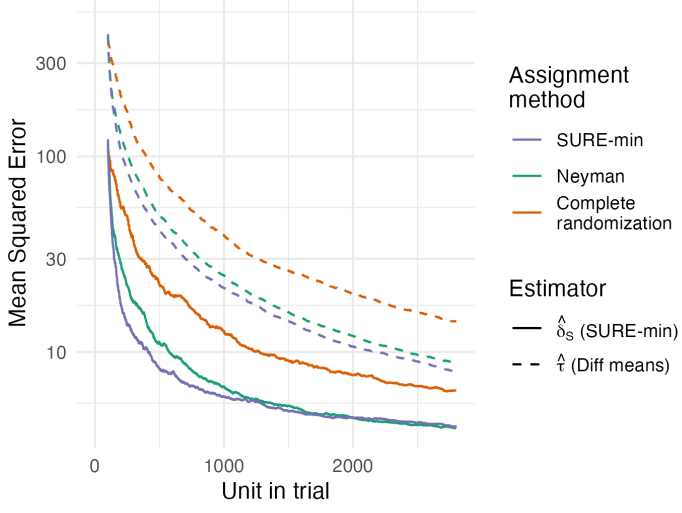
$\longrightarrow$ use Rcpp!

```
81  // [[Rcpp::export]]
82  double calc_stein_min_risk(const arma::vec& tau_hat, const arma::mat& Sigma_hat) {
83    int K = tau_hat.n_elem;
84
85    // Compute xi
86    mat Sigma_sqrt_inv = inv(sqrtmat_sympd(Sigma_hat));
87    vec xi = Sigma_sqrt_inv * tau_hat;
88
89    // Compute matrices
90    mat A = trans(Sigma_hat) * Sigma_hat;
91    mat B = Sigma_hat;
92    double trSigma = trace(Sigma_hat);
93
94    // Eigen decomposition
95    cx_vec eigval;
96    cx_mat eigvec;
97    eig_gen(eigval, eigvec, B);
98    vec D = real(eigval);
99    mat V = real(eigvec);
100
101   // Perform integration
102   Integrand_delta_s integrand(trSigma, A, B, V, xi, D, K);
103   double err_est;
104   int err_code;
105   double integral = integrate(integrand, 0.0, std::numeric_limits<double>::infinity(), err_est, err_code);
106
107   return (1.0 / K) * (trSigma + trSigma * integral);
108 }
```

Simulated Results: Low SNR



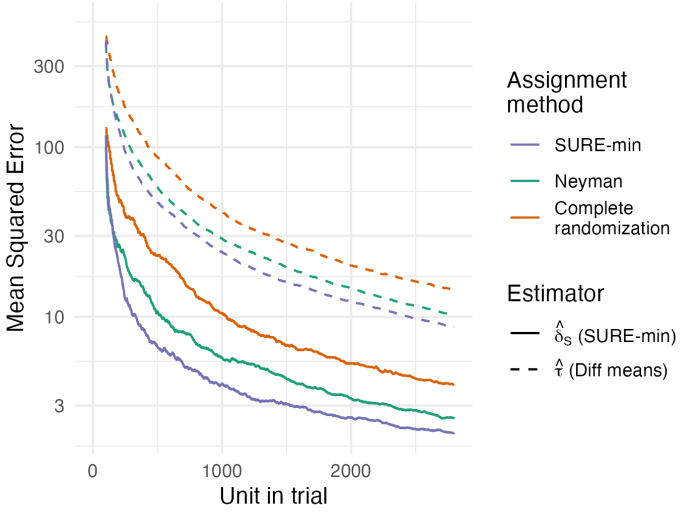
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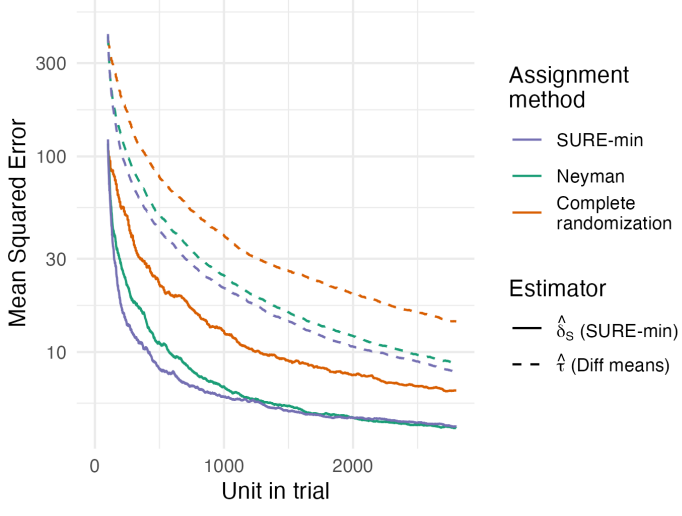
Dense signal; low SNR

- $K = 6$, heteroscedastic
- 2,800 units; first 100 assigned randomly
- Risk trajectories averaged over 1,000 simulations

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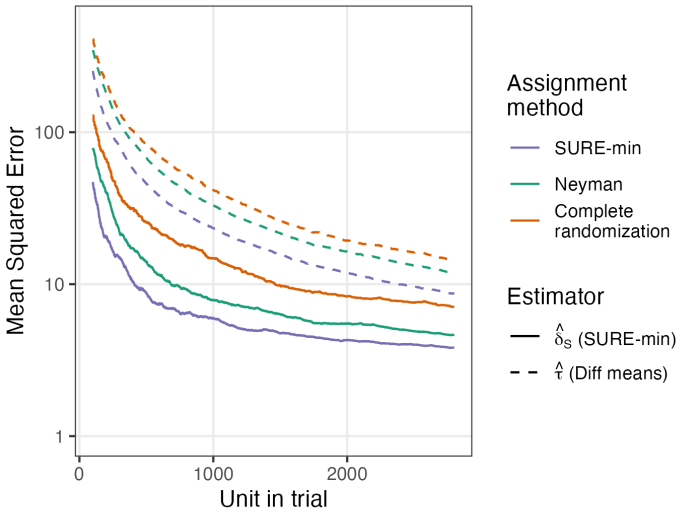


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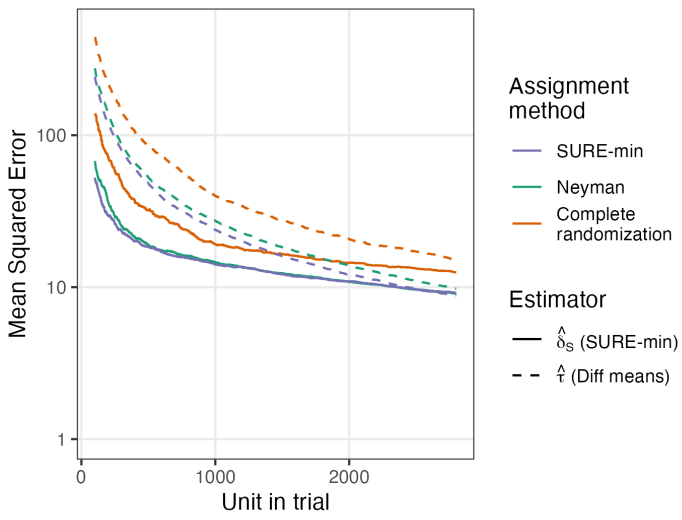
- $K = 6$, heteroscedastic
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- Risk trajectories averaged over 1,000 simulations

- Big gains from using shrinkage estimators
- Adaptivity benefits over Neyman allocation larger when signal is sparse

Simulated Results: High SNR



Sparse signal; high SNR



Dense signal; high SNR

- At high SNR, shrinkage & adaptivity benefits remain for sparse signals
- For dense (many similar-size) signals, benefits are more limited

What about inference?

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- Yes!
  - Adapt "Frequentist Assisted by Bayes" (FAB) intervals from Yu and Hoff (2018)
  - Shorter CIs for the causal effects nonetheless maintain frequentist coverage
  - Intervals account for shrinkage, but are not always-valid nor corrected for adaptivity

Shorter Confidence Intervals

Adapt "FAB" intervals (Yu and Hoff, 2018) to build shorter CIs for the causal effects that nonetheless maintain frequentist coverage.

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SNR	τ setting	Allocation Scheme	Min Wald Coverage	Min FAB Coverage	Avg. FAB length Δ
$\kappa = 0$	–	SURE-min	95%	95%	−15%
		Dimmery	95%	95%	−15%
		Bock	94%	94%	−15%
$\kappa = 4$	sparse	SURE-min	95%	95%	−14%
		Dimmery	95%	94%	−14%
		Bock	93%	93%	−14%
$\kappa = 4$	dense	SURE-min	95%	94%	−12%
		Dimmery	96%	95%	−13%
		Bock	94%	93%	−12%

Δ is mean FAB interval length divided by mean Wald interval length, minus one.

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  - Efficient risk computation via SURE, Laplace transforms, `Rcpp`.
  - Plays a role analogous to Neyman allocation for difference-in-means estimator.
- Risk-minimizing designs consistently allocate more units to the control arm.
- The largest gains occur in the low signal-to-noise regime, where shrinkage is most effective.

Thanks!





Figure 1: Evan T. R. Rosenman (CMC) & Kristen B. Hunter (UNSW)

arXiv: 2602.07404

arXiv draft

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Why use δ_S?

Lemma

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δ_S dominates $\hat{\tau}$ if

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- Testable condition under which estimator has provably lower risk than $\hat{\tau}$ (requires at least $K > 4$)
- Estimator frequently outperforms $\hat{\tau}$ even if condition not met

FAB Intervals: Pratt's results

- Pratt (1963) considered hierarchical model:

$$Y \sim \mathcal{N}(\theta, \sigma^2) \qquad \text{where} \qquad \theta \sim \mathcal{N}(m, \eta^2)$$

- Results depend on idea of constructing *asymmetric intervals*:

$$C_w(y) = \left\{\theta : y + \sigma z_{\alpha(1-w(\theta))} < \theta < y + \sigma z_{1-\alpha w(\theta)}\right\}$$

  - $z_q = \Phi^{-1}(q)$. For any fixed w, coverage is $1 - \alpha$
  - $w \equiv 1/2$ gives the usual Wald interval.

- Under prior $\theta \sim \mathcal{N}(m, \eta^2)$, the optimal rule is

$$w(\theta) = g^{-1}\left(\frac{2\sigma(\theta - m)}{\eta^2}\right), \qquad g(w) = z_{\alpha w} - z_{\alpha(1-w)}.$$

- **Upshot:** rule produces intervals that are shorter near prior's center and longer in the tails. Intervals have $1 - \alpha$ coverage, but are shorter averaged over prior!

FAB Intervals: Our Adaptation

- **Yu & Hoff** (Yu and Hoff, 2018): proposed to *learn* the prior from other groups in the multigroup setting
- Implied prior in our setting (for $\boldsymbol{\delta_B}, \boldsymbol{\delta_S}$) is

$$\boldsymbol{\tau} \sim \mathcal{N}(\mathbf{0}, a\boldsymbol{\Sigma}), \qquad \hat{\boldsymbol{\tau}} \mid \boldsymbol{\tau} \sim \mathcal{N}(\boldsymbol{\tau}, \boldsymbol{\Sigma}), \qquad \mathbb{E}(\boldsymbol{\tau} \mid \hat{\boldsymbol{\tau}}) = \frac{a}{a+1}\hat{\boldsymbol{\tau}}.$$

- **Borrow strength across arms** to estimate the prior scale:

$$\hat{a} = \left(\frac{\hat{\boldsymbol{\tau}}^{\mathsf{T}} \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\tau}}}{K} - 1 \right)_{+}, \qquad \hat{\sigma}_k^2 = \hat{\boldsymbol{\Sigma}}_{kk} = \frac{\hat{V}_0}{n_0} + \frac{\hat{V}_k}{n_k}.$$

- Can then use updated FAB rule using prior variance $a\sigma_k^2$:

$$\hat{w}_k(t) = g^{-1}\left(\frac{2t}{\hat{a}\,\hat{\sigma}_k} \right).$$

- Solve two scalar equations to obtain the interval endpoints.